

GPU Resource Partnership

Aramco SAIL × THAKAA MED

Strategic Collaboration Proposal

Prepared for:

Aramco SAIL

Prepared by:

THAKAA MED

Saudi Arabia's Leading Healthcare AI Company

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Executive Summary

THAKAA MED proposes a strategic GPU resource partnership with Aramco SAIL, leveraging our existing relationship through the Wa'ed investment and DISAI 2025 program. This partnership delivers mutual value: THAKAA MED gains access to GPU compute for healthcare AI training, while SAIL receives expert configuration, knowledge transfer, and ongoing support to maximize infrastructure ROI.

Partnership Overview

Component	Details
SAIL Contribution	<i>1,000 - 2,000 GPU-hours on 4× A40 cluster</i>
THAKAA MED Contribution	<i>Cluster configuration, queue management, knowledge transfer, ongoing support</i>
Relationship Foundation	<i>Wa'ed investment + DISAI 2025 Finalist</i>
Outcome	Optimized SAIL infrastructure ROI + Advanced healthcare AI for Saudi Arabia

1. Why THAKAA MED

Saudi Arabia's Leading Healthcare AI Company

THAKAA MED operates the Kingdom's most comprehensive healthcare AI ecosystem, with production-ready solutions in 3 hospitals. Our AI models deliver measurable clinical impact in radiology, dental, and emergency medicine.

Production AI Portfolio

Product	Application	Clinical Impact
Chest IQ™	<i>Chest X-ray Analysis</i>	<i>90% diagnostic accuracy</i>
Stroke IQ™	<i>3D CT Stroke Detection</i>	<i>Sub-30 second critical alerts</i>
Retina IQ™	<i>Diabetic Retinopathy Screening</i>	<i>Sub-30 second critical alerts</i>
Dental IQ™	<i>Dental Pathology Detection</i>	<i>Live in 3 hospitals</i>

Aramco Ecosystem Relationship

Milestone	Status	Significance
Wa'ed Investment	<i>Active</i>	<i>Aramco's venture arm investing in THAKAA MED</i>
DISAI 2025 Program	<i>Finalist</i>	<i>Aramco SAIL × Qualcomm accelerator program</i>
This Proposal	Strategic Partnership	Deepening collaboration through GPU resources

2. SAIL GPU Infrastructure

Available Resources

Specification	Details
<i>GPU Model</i>	<i>NVIDIA A40</i>
<i>GPU Count</i>	<i>4 GPUs</i>
<i>VRAM per GPU</i>	<i>48 GB</i>
<i>Total VRAM</i>	<i>192 GB</i>
<i>Compute Capability</i>	<i>Professional datacenter GPU</i>
<i>Use Case</i>	<i>AI training, inference, visualization</i>

Current Challenge

GPU infrastructure requires specialized configuration and management to deliver optimal ROI. Without proper queue management and workload orchestration, GPU resources often remain underutilized or inefficiently allocated.

- Manual job scheduling leads to idle GPU time
- Multi-GPU distributed training requires expertise to configure
- Resource monitoring and optimization need dedicated tooling

THAKAA MED Solution

THAKAA MED brings production-tested expertise in GPU cluster management, having configured and operated similar infrastructure for our own healthcare AI workloads. We will transform SAIL's GPU resources into a fully optimized, queue-managed training platform.

3. Partnership Value Exchange

SAIL Provides

Resource	Allocation	Value
GPU Compute Hours	1,500 hours yearly	Training time for healthcare AI experimentation
Infrastructure Access	4× A40 (192 GB)	High-performance training capability
Partnership Duration	12 months (renewable)	Sustained collaboration

THAKAA MED Provides

Deliverable	Description	SAIL Benefit
1. Cluster Configuration	Full setup of queue-based distributed workload system	Production-ready infrastructure
2. ClearML Integration	Job scheduling, prioritization, and monitoring	Automated resource optimization
3. Knowledge Transfer	Training for SAIL team on cluster maintenance	Internal capability building
4. Ongoing Support	Technical assistance and optimization guidance	Maximized infrastructure ROI
5. Documentation	Complete operational runbooks and best practices	Long-term sustainability

Net Outcome

- **SAIL:** Optimized GPU infrastructure with professional management capabilities
- **THAKAA MED:** GPU compute access to advance healthcare AI for Saudi Arabia
- **Kingdom:** Accelerated development of world-class medical AI capabilities

4. THAKAA MED GPU Usage Plan

Training Workloads for SAIL Infrastructure

With 4× A40 GPUs (192 GB total VRAM), THAKAA MED will train selected 2D medical imaging models. The A40 cluster is well-suited for high-resolution 2D workloads, while our larger 3D volumetric training remains on dedicated NVIDIA H200 infrastructure.

Suitable Workloads for A40 Cluster

Workload	Configuration	GPU Hours Estimate
<i>Chest IQ (2D X-ray)</i>	<i>2048×2048, batch 8-16, 4-GPU DDP</i>	<i>200-300 hours</i>
<i>Dental IQ (2D)</i>	<i>1024×1024, batch 16-32, 4-GPU DDP</i>	<i>200-300 hours</i>
<i>Retina IQ (2D fundus)</i>	<i>1024×1024, batch 8-12, 4-GPU DDP</i>	<i>150-250 hours</i>
<i>Model Experimentation</i>	<i>Hyperparameter tuning, architecture tests</i>	<i>200-400 hours</i>
Total Estimated	Selected 2D datasets	750 - 1,250 hours

FOR REFERENCE ONLY: A40 vs H200 Capability Comparison

Workload Type	A40 (48GB × 4)	H200 (141GB × 8)	SAIL Suitability
<i>2D Medical (1024×1024)</i>	<i>Batch 8-16</i>	<i>Batch 16-32</i>	Excellent
<i>2D Medical (2048×2048)</i>	<i>Batch 16-32</i>	<i>Batch 8-16</i>	Limited
<i>3D CT (512³)</i>	<i>Not feasible</i>	<i>Batch 4-8</i>	Not suitable

5. Implementation Plan

Phase 1: Infrastructure Setup (Jan-26 TBD)

- Assess current SAIL GPU cluster configuration
- Install and configure ClearML server for job orchestration
- Set up queue management with priority levels
- Configure multi-GPU distributed training (4-way DDP)
- Establish monitoring and resource tracking dashboards

Phase 2: Knowledge Transfer (Feb-26 pre Ramadan)

- Training sessions for SAIL team on cluster management
- Documentation handover: operational runbooks, troubleshooting guides
- Best practices for GPU utilization and job scheduling
- Hands-on practice with supervised job submission

Phase 3: Production Usage + collab announcement

- THAKAA MED submits training jobs through queue system
- Monthly utilization reports shared with SAIL
- Ongoing optimization recommendations
- Technical support for any infrastructure issues

6. Summary

A Partnership That Benefits Everyone

Stakeholder	Receives	Provides
Aramco SAIL	<i>Optimized infrastructure, knowledge transfer, ongoing support</i>	<i>1,000-2,000 GPU-hours</i>
THAKAA MED	<i>GPU compute for healthcare AI training</i>	<i>Configuration, training, documentation, support</i>
Saudi Arabia	Advanced healthcare AI + Stronger AI ecosystem	Vision 2030 alignment

Why This Partnership Makes Sense

1. Existing Relationship: Wa'ed investment and DISAI 2025 finalist status establish trust and alignment
2. Mutual Value: Both parties receive tangible benefits that advance their core missions
3. Low Risk: THAKAA MED has proven expertise; SAIL retains full infrastructure ownership
4. Kingdom Impact: Accelerates Saudi Arabia's healthcare AI leadership

Proposed Next Steps

5. Review and feedback on this proposal
6. Technical scoping call to assess current SAIL infrastructure
7. Agreement on GPU-hour allocation and timeline
8. Kick-off Phase 1 implementation

Together, we advance Saudi Arabia's healthcare AI capabilities.

Submitted by THAKAA MED
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